

AREA SCIENCE PARK
RESEARCH AND TECHNOLOGY INSTITUTE
LABORATORY OF DATA ENGINEERING

Web Application for DPCfam and DPCstruct Data Exploration

Report 9 (10 - 16 March 2026) : Web Development
(DPCStruct Backend)

Emmanuel Nyandu Kagarabi
Master in Data Management and Curation (MDMC - SISSA)

Supervisors :
Valerio Piomponi
Elaheh Saadat

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1 Introduction

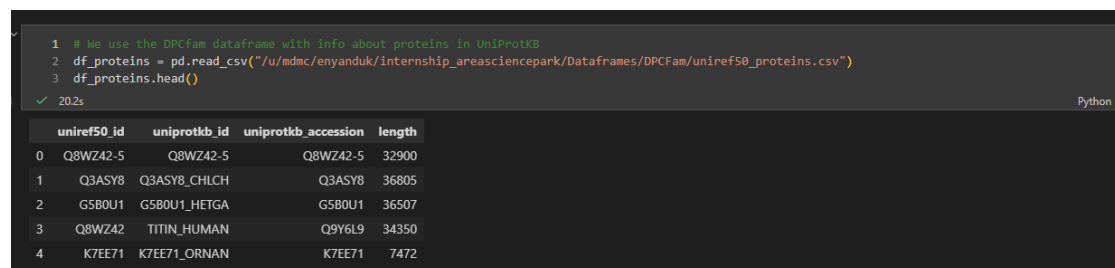
This report is the ninth in a sequence documenting the weekly assigned tasks of our internship. Focusing on the DPCStruct dataset ([Barone et al., 2025](#)), we present the progress made on the dpcstruct django-app, where we literally added CATH/SCOP Annotations. Likewise, the global project was updated so that users can choose any database source among the four available (DPCfam, DPCstruct, Pfam, and Uniprot) from the landing page, enter a valid ID, and then, if everything goes well, display everything we wanted them to explore, except the internal 3D Viewer for DPCstruct representatives.

2 Development

2.1 Fetching UniProt Metadata

From Report 8, we pointed out the fact that we weren't given protein length data for dpcstruct, which is crucial for displaying the graphical representation highlighting the overlap between a given protein ID and matching samples in DPCfam, DPCstruct, and Pfam.

In fact, for DPCstruct, we have **1, 642,061** domains distributed over **1,252,179** proteins, organised into **28,246** metaclusters. **25%** of these domains already existed in our dataframe derived from UniRef50 annotated (Figure 1):



```
1 # We use the DPCfam dataframe with info about proteins in UniProtKB
2 df_proteins = pd.read_csv("/u/mdmc/enyanduk/internship_areasciencepark/Dataframes/DPCFam/uniref50_proteins.csv")
3 df_proteins.head()
```

	uniref50_id	uniprotkb_id	uniprotkb_accession	length
0	Q8WZ42-5	Q8WZ42-5	Q8WZ42-5	32900
1	Q3ASY8	Q3ASY8_CHLCH	Q3ASY8	36805
2	G5B0U1	G5B0U1_HETGA	G5B0U1	36507
3	Q8WZ42	TITIN_HUMAN	Q9Y6L9	34350
4	K7EE71	K7EE71_ORNAN	K7EE71	7472

Figure 1: UniRef50_annotated : **23M** samples.

By filtering DPCstruct domains in Figure 1, we were left with 75% of proteins without information on their length. Taking inspiration from the [UniProt ID Mapping](#) page, our script ([fetch_uniprot_complete.py](#)) could help fetch the missing lengths in batches.

However, given the fact that some proteins are currently being moved within UniProt, from **UniProtKB** to **UniParc**, our fetching operation was unsuccessful after a couple of iterations (7) for some unknown reasons.

Therefore, we decided to download sequences from UniProt and then, for any existing protein ID, it should be manageable to obtain its length .

What release should we download, then?

Optimally, it would be one corresponding to the publication of [Barone et al. \(2025\)](#). Unfortunately, file sizes were problematic; we downloaded the latest release:

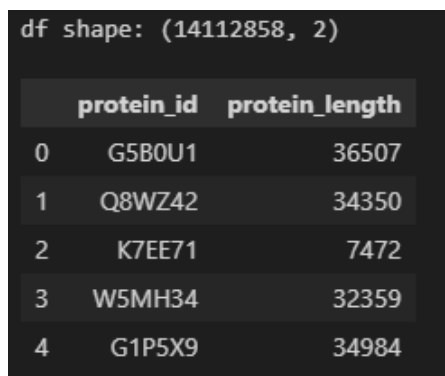
```
wget -c https://ftp.uniprot.org/pub/databases/uniprot/current_release/knowledgebase/complete/uniprot_sprot.fasta.gz
wget -c https://ftp.uniprot.org/pub/databases/uniprot/current_release/knowledgebase/complete/uniprot_trembl.fasta.gz
```

The first command corresponds to **Swiss-Prot** (manually reviewed, high-quality entries, ≈ 90 MB) data and the second to **TrEMBL** (automatically annotated, much larger, ≈ 55 GB).

By running the `extract_lengths_from_fasta.py` script on the two downloaded files, we retrieved and saved the extracted lengths. For the remaining proteins ($\approx 30\%$ of the total), the `fetch_missing_lengths.py` script gathered their lengths, through a systematic check in both **UniProtKB** and **UniParc**.

Hopefully, we obtained all missing information on DPCStruct domains and preprocessed them using the `dpcstruct_all_proteins_lengths.ipynb` notebook.

Altogether, we concatenated information on proteins from DPCfam, Pfam Annotated and DPCstruct, into a single dataframe (Figure 2):



df shape: (14112858, 2)		
	protein_id	protein_length
0	G5B0U1	36507
1	Q8WZ42	34350
2	K7EE71	7472
3	W5MH34	32359
4	G1P5X9	34984

Figure 2: UniProt Metadata: 14M samples. Check out `dpc_protein_lengths.ipynb`

Likewise, all unique Pfam IDs (Families or clans) were collected from (DPCfam, Pfam, DPCstruct) annotated dataframes and were grouped into a consistent one to be referenced in both DPCfam and DPCstruct (Figure 3):

```
df shape: (19352, 2)
Unique PF families: 18734
Unique CL clans: 617
Unknown: 1
```

	pfam_id	pfam_type
0	PF07679	Family
1	PF02818	Family
2	PF18362	Family
3	PF00069	Family
4	PF00041	Family
5	PF09042	Family
6	PF13927	Family
7	PF00047	Family
8	PF13018	Family
9	PF12951	Family

Figure 3: PFAm Data Source : 19.3m. Check out [dpc_pfam_ids.ipynb](#).

2.1.1 Project Updates

Moving on, I completed the preprocessing of CATH and SCOP data, using the [dpcstruct_explore_cath_in_depth.ipynb](#) and [dpcstruct_explore_scop_in_depth.ipynb](#) notebooks:

scop_query	dpc_mid	dpc_target	q_range	t_range	qlen	tlen	qcov	tcov	qliden	qlmscore	tlmscore	almscore	ldlt	pidant
0	1.10.3930	MC30	ADA7Z7D914_50	1-78	64-141	78	150	1.00	0.52	0.52	1.00	0.74		
1	1.10.45	MC79	ADA512D188_75	2-38	26-62	38	62	0.97	0.60	0.60	0.97	0.60		
2	3.30.280	MC79	ADA439LE1_79	1-100	1-100	100	104	1.00	0.96	0.96	1.00	0.96		
3	3.30.1590	MC31	ADA7Y3MCQ8_93	1-103	1-114	143	202	0.90	0.58	0.58	0.90	0.58		
4	2.20.20	MC31	ADA7Y3MCQ8_95	2-308	400-600	231	601	0.82	0.33	0.33	0.82	0.33		
5	3.30.230	MC35	ADA39AC3Q8_95	7-262	2-344	263	637	0.97	0.38	0.38	0.97	0.38		
6	3.40.228	MC124	ADA7V7MC10_124	3-103	13-140	128	223	0.96	0.57	0.57	0.96	0.57		
7	1.10.155	MC124	ADA7V7MC10_124	7-75	2-71	75	266	0.92	0.26	0.26	0.92	0.26		
8	3.30.1710	MC124	ADA8121M6_124	19-91	114-198	91	198	0.80	0.43	0.43	0.80	0.43		
9	1.10.4880	MC212	ADA3M18EV_212	3-344	2-342	337	340	0.96	0.99	0.99	0.96	0.99		

(a) CATH

scop_query	dpc_mid	dpc_target	q_range	t_range	qlen	tlen	qcov	tcov	qliden	qlmscore	tlmscore	almscore	ldlt	pidant
0	Q2YK02_0001190	MC79	ADA3E259L_79	1-99	1-99	100	99	0.99	1.00	0.99	0.98	0.99	0.99	55.5
1	Q2YK02_000088	MC124	ADA7V7MC10_124	1-103	5-107	103	101	1.00	0.96	0.96	1.00	0.96	0.96	10.9
2	Q2YK02_000073	MC124	ADA7V7MC10_124	23-216	1-192	216	192	0.89	1.00	0.99	0.89	0.99	0.99	13.3
3	Q2YK02_0001534	MC138	H0E529_138	39-231	1-192	240	206	0.80	0.93	0.93	0.80	0.93	0.93	16.0
4	Q2YK02_0000810	MC212	ADA7V7MC10_212	53-295	1-256	300	257	0.81	1.00	0.99	0.77	0.89	0.89	31.1
5	Q2YK02_0001018	MC212	ADA7V7MC10_212	25-161	1-134	163	134	0.82	1.00	0.99	0.81	0.99	0.99	70.1
6	Q2YK02_0001118	MC295	ADA7V7MC10_295	2-81	12-102	81	102	0.99	0.93	0.93	0.99	0.93	0.93	26.8
7	Q2YK02_0000604	MC353	ADA5480V3_353	1-25	84-108	25	102	1.00	0.16	0.16	0.99	0.16	0.16	52.0
8	Q2YK02_0000397	MC353	ADA7V7MC10_353	19-138	1-120	148	121	0.81	0.99	0.99	0.79	0.96	0.96	35.8
9	Q2YK02_0001018	MC212	ADA7V7MC10_212	2-110	3-100	110	101	0.94	0.97	0.97	0.94	0.97	0.97	13.1

(b) SCOP

Figure 4: Comparison with CATH and SCOP.

Likewise, I updated DPCfam Metaclusters by merging information from DPCFam Standard and DPCFam B ([dpcfam_all_mcs_props.ipynb](#) and [dpcfam_all_mcs_sequences.ipynb](#) notebooks), producing Figure 5.

	mcid	size_uniref50	avg_len	std_avg_len	lc_percent	cc_percent	dis_percent	tm	pfam_da	da_percent	avg_ov_percent	overlap_label
0	MC1	17931	185.68	28.77	4.72	0.00	18.44	0.01	PF13614	44.23	80.82	Equivalent
1	MC3	28	215.14	25.44	5.08	0.38	30.15	0.00	PF14631	8571.43	24.57	Shifted
2	MC4	617	59.91	6.07	4.99	0.00	1.87	1.26	PF03600	62.84	7.54	Shifted
3	MC5	26	253.19	23.11	4.88	0.00	7.51	0.00	PF12937	6800.00	16.89	Extended
4	MC7	32	65.53	3.30	3.39	0.00	0.10	0.00	PF01243	7407.41	6.27	Shifted

Figure 5: DPCFam Metaclusters: I dropped the column PFam Sequences, which was missing in DPCFamB, since I felt it wasn't instrumental.

2.2 Routine and Results

On one side of the coin, each preprocessed CSV file was mapped to a table in **PostgreSQL**, and, if relevant, connected to an existing one, and its indexes were defined for optimising the query-response process.

On the other side of the coin, in **Django**, the **Model-View-Template** pattern was created/updated accordingly, as well as the definition of URLs.

Putting everything together :

1. Data from Figure 5 was used to update the `DpcfamMcsProperty` model, which, together with all dependent files (filter, table, view, template, url) resulted in Figure 6:

DPCFam Metaclusters

Enter MCID:

MCID	Size UniRef50	Avg. Len.	Std avg len	% LC	% CC	% DIS	Avg. TM	Pfam DA	% DA	% Avg. Ov.	Overlap Label
MC1	17931	185.68	28.77	4.72	0.0	18.44	0.01	PF13614	44.23	80.82	Equivalent
MC3	28	215.14	25.44	5.08	0.38	30.15	0.0	PF14631	8571.43	24.57	Shifted
MC4	617	59.91	6.07	4.99	0.0	1.87	1.26	PF03600	62.84	7.54	Shifted
MC5	26	253.19	23.11	4.88	0.0	7.51	0.0	PF12937	6800.0	16.89	Extended
MC7	32	65.53	3.3	3.39	0.0	0.1	0.0	PF01243	7407.41	6.27	Shifted
MC11	32	134.97	11.99	2.05	0.0	15.61	0.0	UNKNOWN	10000.0	0.0	Shifted
MC15	139	81.21	5.05	4.96	0.18	13.81	0.03	UNKNOWN	0.0	0.0	None
MC17	26	243.69	25.93	3.44	0.0	31.88	0.0	PF08590	9000.0	5.31	Shifted
MC18	41	143.29	8.57	3.65	0.0	1.12	0.13	PF18736	9642.86	94.74	Equivalent
MC19	120	71.57	7.7	8.85	2.25	4.86	1.69	PF11915	94.07	13.66	Shifted

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Figure 6: DPCFam Metaclusters Page.

2. Data presented in Figure 4 was used to create the `DpcStructCath` and `DpcStructSCop` models, which, associated with their respective files (filter, table, views, template, urls) direct the user to a page displayed in Figure 7:

DPCStruct Metaclusters

Properties

CATH

SCOP

Search MCID:

Search Fold:

Search

Current View: CATH Fold Annotations

CATH Fold	MCID	DPC Target	Q.Range	T.Range	Q.Cov.	T.Cov.	Q.TM-Score	T.TM-Score	A.TM-Score	DDT	% Ident.
3.30.2160	MC9905	A0A7H4MW51_9905	3-73	12-76	0.91	0.47	0.26	0.17	0.17	0.51	9.5
1.20.1580	MC9630	C2NRS3_9630	1-132	78-209	0.95	0.63	0.93	0.62	0.62	0.87	50.0
1.10.3990	MC957	A0A521GNK4_957	7-95	2-84	0.82	0.98	0.56	0.69	0.69	0.55	24.4
3.90.1140	MC9528	A0A816R5E1_9528	1-180	13-193	0.99	0.94	0.97	0.91	0.91	0.95	87.2
3.90.1740	MC9528	A0A7S0DPV3_9528	6-201	10-135	0.95	0.52	0.5	0.43	0.43	0.51	11.2
2.20.80	MC95	A0A7R9DN90_95	2-208	400-600	0.82	0.33	0.58	0.26	0.26	0.65	13.1
2.30.230	MC95	A0A3B4CDQ0_95	7-262	2-244	0.97	0.38	0.86	0.36	0.36	0.78	33.2
1.20.1520	MC9337	A0A402F1A5_9337	1-120	1-120	0.96	0.99	0.95	0.98	0.98	0.98	86.6
1.10.555	MC931	A0A1V9Z9Y6_931	5-199	79-302	0.93	0.74	0.61	0.44	0.44	0.52	13.5
3.30.1560	MC93	A0A7Y5MQG8_93	1-128	1-114	0.9	0.56	0.6	0.45	0.45	0.61	11.6

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Figure 7: Metaclusters/DPCStruct Page. By default, it displays MCs Properties with clickable links, but, if desired, the user can also choose CATH or SCOP Annotations and filter the paginated table by either an exact MCID or CATH/SCOP fold ID.

- Data from Figure 3 was mapped to theDpcPfamDomain model, validating PFam ID queries, and then facilitating the display of matching metaclusters from DPCfam (using the average overlap) and DPCstruct(using the pfam score), as shown in Figure 8:

localhost:8000/pfam/PF03243/

DPCFam & DPCStruct Unified Database

Home

About

Metaclusters

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FAQs

Pfam Domain: PF03243

Metaclusters containing this Pfam domain

Visit PF03243 in InterPro Database

6 Metaclusters Found

DPCFam (Avg. Overlap: 3 Metaclusters)

Rank	MCID	Overlap Score
1	MC21651	68.7%
2	MC33220	44.7%
3	MC388534	24.3%

DPCStruct (Pfam Score: 3 Metaclusters)

Rank	MCID	Pfam Score
1	MC16858	100.00
2	MC40874	61.11
3	MC45067	77.78

Back to Search

Figure 8: PFam matches in both DPCfam and DPCstruct.

4. Data from Figure 2 was used to create the `DpcUniprotProtein` model, which validates queries coming from the UniProt database, and then ultimately facilitates the display of matching samples (DPCFam, DPCStruct, Pfam), for a given protein ID, resulting in Figure 9:

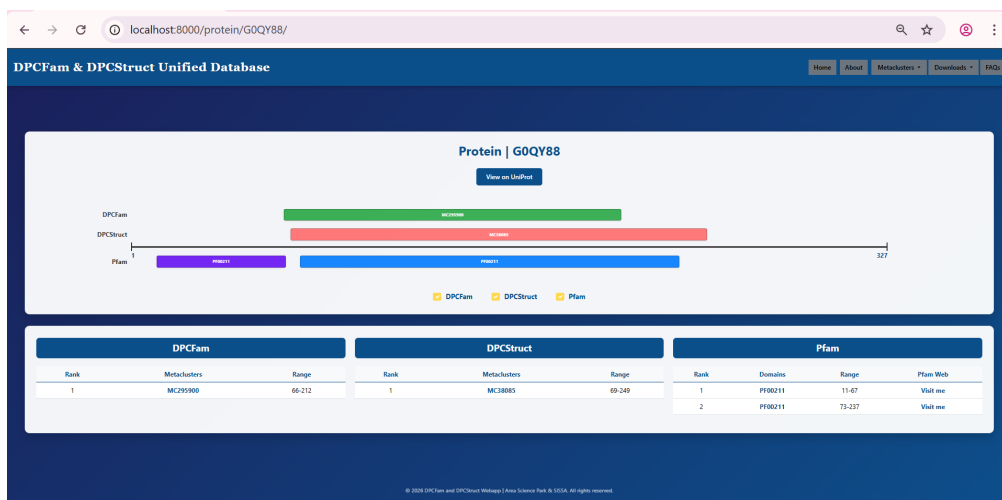


Figure 9: Matching samples in DPCFam, DPCStruct and Pfam.

3 Conclusion

In order to display the current state of the project, please check out the [README.md](#) file. CSV files are available in [Google Drive](#). I am currently exploring the remaining files (representative PDBs) in order to render the 3D viewer using the [Mol*](#) toolkit, our ending point.

References

- Barone, F., Laio, A., Punta, M., Cozzini, S., Ansuini, A., and Cazzaniga, A. (2025). Unsupervised domain classification of alphafold2-predicted protein structures. *PRX Life*, 3(2):023009.